Project Compass

Mission Statement (problem, stakeholders, outcomes and benefits): Compass aims to address the challenge of managing daily tasks and appointments for busy individuals by offering an intuitive scheduling assistant that optimizes time management and boosts productivity. By leveraging advanced AI technology, we deliver personalized and efficient planning solutions that cater to the dynamic needs of our users. Our mission is to transform the way our stakeholders navigate their day, resulting in tangible benefits such as enhanced efficiency, reduced stress, and more meaningful use of time.

Target audience: General public - those who want to be more efficient with their time.

Functional Requirements

1.1 All registered users must log in to their existing account.

1.1.1 The user can choose their preferred login method

1.1.1.1 The user can sign in with Google.

1.1.1.2 The user can manually enter username/email and password.

1.1.1.3 If the user does not have an existing account, can sign up with Google or manually.

1.1.2 If the user forgets the password, the user can click on “Forget password?”.

1.1.3 The system must validate the username and password.

1.3.1.1 If the credentials are valid, the user will be redirected to the web page.

1.3.1.2 If the credentials are invalid, the user can retry logging in for a maximum of 3 times. The account will be locked out after 3 tries.

1.2 For first time login users, the system can gather the user’s timetable inputs.

1.2.1 The user can input their timetable in different formats

1.2.1.1 The user can upload images of their schedules.

1.2.1.2 The user can upload their schedule in a standardized format (.cal)

1.2.1.3 The user can grant permission to the system to import the Google Calendar.

1.2.2 The web system will consolidate the inputs from the user and update the web calendar.

1.3 For subsequent logins into the web system, the users can view a live dashboard.

1.3.1 The user can toggle their timetable view between monthly and weekly format.

1.3.2 The user can view the list of priorities ordered by its assigned priority number.

1.4 The system can allow users to add events into the timetable (which may be filled or empty).

1.4.1 The user must enter details of the event.

1.4.1.1 The user must describe the event within 50 words.

1.4.1.2 The user must provide the event's start and end time in the 24 hour format.

1.4.1.3 The user must assign a priority number to each event added.

1.4.1.4 The user must provide duration of the event, a minimum of 5 minutes for each slot.

1.4.2 The system will allocate a suitable slot and update in the timetable.

1.4.3 The system must allocate sufficient traveling time between each event.

1.4.3.1 The user must be connected to the Internet and enable GPS when allocating the event to access Google Maps.

1.4.4 If there is a clash of events, the system will notify the user.

1.4.4.1 The system will request for the user's permission to reschedule or remove the older event.

1.4.4.2 If the user has rescheduled or removed the older event, the system will update the timetable automatically.

1.4.5 The system synchronizes the changes in events in the timetable on Google Calendar.

1.5 The system can support timetable access in different formats/ on other applications.

1.5.1 The user can export the timetable into PDF or JPG format to be printed out.

1.6 The system can allow multiple users to share their timetables.

1.3.1 Users must be connected to WIFI before they can share timetables with each other.

1.3.2 The system can help to find common vacant time slots between timetables.

Non-functional requirements

<u>Security</u>	User must be authenticated before allowed access to the schedules Personal schedules will only be visible to the user and a select group of other trusted users, calendar events can be individually tagged as public or private
<u>Reliability</u>	Upon opening the web page, the full functionality should be restored within **5** seconds

Performance	Upon receiving a screenshot of a timetable, the webpage should output a schedule within 15 seconds
Maintainability	Upon system error / website crash, must come back online within 3 hours
Scalability	The system must be able to support up to 5 persons to find common time slot The system will be able to support at least 1000 users to access the webpage concurrently To start, the system will be able to accept 40,000 unique user accounts, each of which will be allocated a maximum of 1MB for up to 6 months worth of calendar events.
Usability	The system should be able to export the created schedule to the user's calendar in PDF/JPG
Regulatory Compliance	The system will ensure compliance with relevant laws, regulations, and industry standards related to data privacy, security, and accessibility In a local context, the system will be designed to be compliant with Singapore's PDPA. Users will be asked for consent to store Personal calendar data for our scheduling service. Data will be retained as long as it is relevant for service delivery, which will be six months as that is as long as users will be allowed to upload. Furthermore, data used for business purposes such as AI model training will be properly anonymised before usage
Integration	The webpage can seamlessly integrate with external systems, APIs, or third-party services (e.g., Google Calendar).(syncing with google calendar)
Compatibility	Users can access the webpage on different devices, operating systems and web browsers to maximize reach and usability